

Network Traffic Analysis - DNS Resolution Failure Investigation

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Lab Setup

This lab was set up using AI to generate a packet capture file. This file was copied into an Ubuntu VM environment in order to utilize tcpdump for analysis.

Scenario

A workstation user reports that **some network resources are failing to load while other network activity appears to continue working**. Network traffic was captured from the affected workstation during the problem. **Determine what the packet evidence tells you about the failure.**

Tcpdump return: (tcpdump -r /home/aaron/case_02_capture.pcap)

Observations

1. The capture contains approximately five seconds of activity, all packets occur between 15:15:00 and 15:15:05
2. IP 10.10.20.15 makes a DNS request to 10.10.20.53 for [example.com](#), and the IP returned is 93.184.216.34
3. TCP connection is established between 10.10.20.15 and 93.184.216.34 through a three way handshake
4. 10.10.20.15 sends an HTTP GET request (for root/default resource) to 93.184.216.34 and receives an HTTP 200 OK response.
5. 10.10.20.15 sends an ICMP echo request to 10.10.20.1 and receives an ICMP echo reply.

6. 10.10.20.15 makes a DNS request to 10.10.20.53 for the IP associated with portal.internal, 10.10.20.53 returns an ICMP error message that identifies the UDP port 53 is unreachable.

7. 10.10.20.15 makes a DNS request to 10.10.20.53 for the IP associated with updates.internal, 10.10.20.53 again returns an ICMP error message that identifies the UDP port 53 is unreachable.

8. 10.10.20.15 makes a DNS request to 10.10.20.53 for the IP associated with [status.example.net](#). The DNS request is successful and the IP 198.51.100.27 is returned.

9. TCP connection is established between 10.10.20.15 and 198.51.100.27 through a three way handshake.

IP Address	Role	Evidence
10.10.20.15	Affected workstation	Initiates DNS queries, initiates web connections, initiates queries that return errors.
10.10.20.53	DNS server	This is the IP that DNS requests are sent to and returned from.
93.184.216.34	Web server for example.com	Workstation successfully established a TCP connection and received an HTTP 200 OK response.
10.10.20.1	Unknown network host	An ICMP echo was sent from 10.10.20.15 and returned confirming this host is reachable.
198.51.100.27	Web server for status.example.net	Workstation completed TCP three way handshake on HTTPS port 443.

Findings

1. Basic connectivity is working. The workstation can communicate with other hosts.
2. DNS is partially successful, some queries receive valid responses, while others receive ICMP port unreachable errors.
3. The underlying cause is unknown. It cannot be determined from this capture alone whether the result of the errors stems from a DNS service issue, firewall configuration, or another problem.

DNS requests made to the domain names portal.internal and updates.internal are failing. (17410, 17411) The DNS requests for both domains return an ICMP error message that identifies UDP port 53 as unreachable. This confirms that these DNS queries failed but does not reveal the root cause. This failure can impact the workstation by compromising availability and limiting access to resources necessary for business continuity. We can determine the specific domains where DNS succeeds and fails, we can also determine that the failure occurs at the DNS level because of the error message reporting that UDP port 53 is unreachable. We know the workstation is capable of establishing a TCP connection with hosts.

Recommendation

My recommendation for taking steps toward remediation is first investigating whether the DNS service is functioning properly at all times. Examine the DNS server's logs from 15:15:02-15:15:04, when the ICMP errors occurred. We know the workstation is capable of establishing a connection to hosts, but we cannot confirm from this capture whether DNS availability remains consistent at all times. A packet capture over a longer period of time could reveal more. It should also be confirmed that the firewall is not blocking traffic to UDP port 53 or misconfigured in another way that could cause DNS failures.

Conclusion

This investigation used tcpdump to analyze network traffic from a workstation experiencing connectivity issues. The capture analysis showed that basic network connectivity was functioning, while certain DNS requests failed with ICMP port unreachable errors. The root cause could not be determined from the available capture, but the evidence helped determine the issue as a DNS related communication failure. This project helped develop my ability to interpret network traffic using tcpdump, recognize DNS queries and TCP connections, identify ICMP port unreachable errors, and strengthened my understanding of network communication failures.